Addressing the Explainability and Safety Against Adversarial Attacks of the ReneeXML

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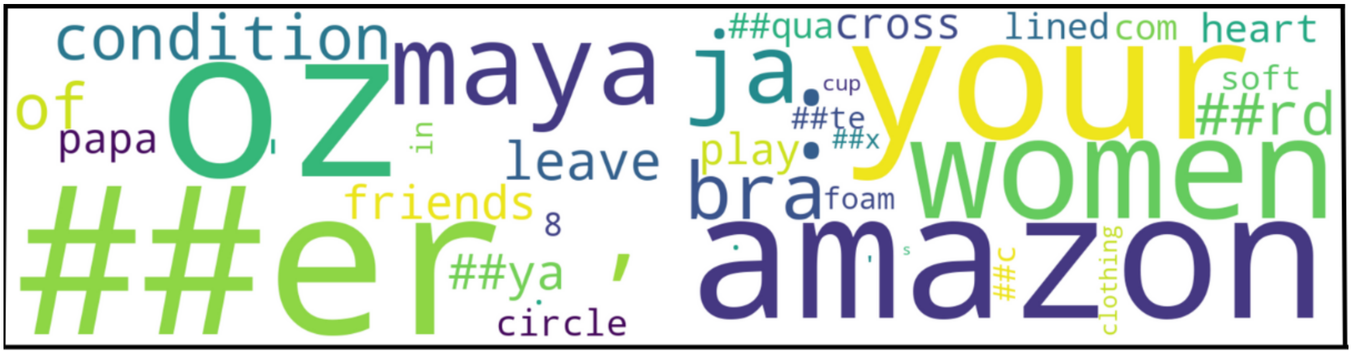


Figure 1. This word cloud reflects the average importance of tokens across multiple instances analyzed using Local Interpretable Model-agnostic Explanations.

1 Abstract

Extreme Multi-label Classification (XML) tasks require models to assign relevant labels from a potentially massive label space, posing significant challenges in scalability, interpretability, and security. Building upon the RENEE framework—a state-of-the-art end-to-end trainable XML model—we present a unified approach that enhances both model explainability and robustness against adversarial attacks. Our framework extends RENEE by integrating a token-level adaptation of LIME, enabling detailed local interpretability through perturbation analysis, consistency checks, and comprehensive visual diagnostics. Simultaneously, we bolster model resilience by implementing white-box adversarial attacks—including FGSM, PGD, MIM, and DeepFool—applied at the embedding level, and introduce an adversarial training regimen that incorporates a balanced loss formulation. Additionally, gradient-based saliency methods are employed to further elucidate decision rationales, particularly under adversarial conditions.

Evaluations on the LF-AmazonTitles-131K dataset demonstrate that our approach preserves high predictive performance while significantly enhancing model dependability. This work contributes a robust, interpretable pipeline for XML, bridging the gap between high-accuracy predictions and the essential need for trustworthy, defensible machine learning systems.

2 Objectives

Extreme Multi-label Classification (XML) involves assigning a subset of relevant labels from a very large label space, often containing hundreds of thousands of classes. Traditional XML methods typically rely on multi-stage pipelines or sampling-based techniques, which can hinder efficiency and obscure model decision-making.

In this project, we extend the state-of-the-art RENEE framework—a scalable, end-to-end XML model—by integrating two key enhancements:

2.1 Explainability through LIME Adaptation:

We adapt the LIME methodology to work at the token level with the RENEE model. By generating perturbed input texts and fitting a weighted linear surrogate model, our framework elucidates which tokens most influence each prediction. Visual tools like bar charts, heatmaps, and word clouds, alongside consistency checks, ensure robust and interpretable explanations.

2.2 Robustness via Adversarial Defenses:

We enhance the model's resilience by implementing white-box adversarial attacks (FGSM, PGD, MIM, DeepFool) at the embedding level and incorporating adversarial training. This mixed training approach, paired with gradient-based saliency maps, not only defends against adversarial perturbations but also provides insights into how such attacks affect model predictions.

Combined, these enhancements yield a reliable XML pipeline that maintains high accuracy on clean data, demonstrates improved resilience against adversarial manipulations, and offers clear insights into its decision process—paving the way for safer and more transparent real-world applications.

3 Existing Renee:Deep Learning Framework for Extreme Multi-Label Learning

3.1 Overview

The RENEE framework introduces an efficient and scalable approach to Extreme Multi-label Classification (XML), where the number of possible labels can be in the hundreds of thousands or more. Unlike traditional methods that use decoupled pipelines or sampling-based heuristics, RENEE demonstrates that end-to-end deep learning is both feasible and advantageous at this scale through a set of architectural innovations and system-level optimizations.

3.2 Core Architecture

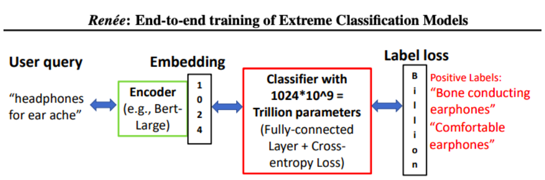


Figure 2. Renee Model with 1 Billion Classes

3.2.1 Modular Design.

- Built on PyTorch, RENEE is structured around the `GenericModel` class using `nn.Sequential` for modular extensibility.
- Supports hybrid parallelism (data + model), making it suitable for distributed setups.
- Enables easy integration of custom encoder and classifier modules.

3.2.2 Encoder Module.

- Utilizes transformer-based models such as BERT or RoBERTa for text embedding.
- Operates in data-parallel mode across GPUs for efficient batch processing.
- Produces dense representations that feed into the classifier.

3.2.3 Classifier Module.

- Maps encoder outputs to extremely large label spaces.
- Employs a custom fully connected (xfc) layer with label sharding across GPUs.
- Uses parameters like `numy_per_gpu` for flexible partitioning.

3.3 Training Optimizations

3.3.1 Loss and Backpropagation Enhancements.

- **Skip Loss**: Bypasses full loss tensor computation using fused CUDA kernels for direct gradient computation.
- **Sparse Loss**: Computes loss only over positive labels, significantly reducing memory and compute.
- **Bottleneck Layer**: Compresses encoder output dimensionality (e.g., from 1024 to 64) to reduce parameter count.

3.3.2 Memory and Speed Optimizations.

- **Gradient Accumulation**: Simulates large batch sizes via micro-batch updates, useful for low-memory environments.
- **Mixed-Precision Training**: Utilizes FP16 via NVIDIA Apex to reduce memory usage and speed up training.
- **Communication Overlap**: Performs asynchronous gradient synchronization to parallelize communication and computation.

3.3.3 Custom CUDA Integration.

- Uses custom CUDA kernel (`xfc_gemm_cuda`) for optimized forward and backward passes in the classifier.
- Gracefully falls back to PyTorch autograd if CUDA is unavailable.

3.4 Inference and Evaluation

3.4.1 Efficient Inference.

- Uses Approximate Nearest Neighbor (ANN) methods like DiskANN for fast retrieval of top-k labels.
- Reduces inference time significantly with negligible accuracy loss.

3.4.2 Evaluation Pipeline.

- The `FullPredictor` class aggregates predictions and computes metrics such as Precision@k and nDCG@k.
- Supports evaluation logging in formats like `.npz` and `.tsv`.
- `PreTokBertDataset` and `XCCollator` handle memory-efficient data loading and batching.

3.5 Relevance to Our Project

3.5.1 Suitability for Dependable XML.

- RENEE supports massive label spaces and distributed training, making it suitable for extreme-scale applications.
- Its modular design allows integration of new layers and techniques for dependability without disrupting the core pipeline.

3.5.2 Our Enhancements.

- **Explainability**: Introduced a token-level LIME adaptation to interpret model decisions; supported with bar plots, word clouds, and consistency metrics.
- **Robustness**: Integrated white-box adversarial attacks (FGSM, PGD, MIM, DeepFool) at the embedding level and adversarial training using a weighted loss strategy.
- Incorporated gradient-based saliency maps to visualize model attention shifts under attack conditions.

4 Adversarial Attacks: Theory and Implementation

4.1 Taxonomy of Attacks

Adversarial attacks aim to deceive machine learning models through carefully crafted perturbations. We categorize the implemented attacks as follows:

- **White-Box**: Attacker has full model access (FGSM, PGD, MIM)
- **Decision-Based**: Relies on model outputs (DeepFool)
- **L_p-Bounded**: Constrains perturbation magnitude (FGSM, PGD, MIM)

4.2 Attack Methodologies

4.2.1 Fast Gradient Sign Method (FGSM).

FGSM generates adversarial examples using the gradient of the loss function:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y)) \tag{1}$$

where ϵ controls perturbation magnitude. This single-step attack prioritizes computational efficiency over stealth.

4.2.2 Projected Gradient Descent (PGD).

PGD extends FGSM with iterative refinement and projection:

$$x_{adv}^{t+1} = \text{Proj}_\epsilon \left(x_{adv}^t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(x_{adv}^t, y)) \right) \tag{2}$$

The projection operator Proj_ϵ ensures perturbations remain within an L_∞-ball of radius ϵ. PGD is considered the "gold standard" for white-box attacks.

4.2.3 DeepFool (DF).

This decision-based attack computes minimal perturbations to cross decision boundaries:

$$\min_\delta \|\delta\|_2 \quad \text{s.t.} \quad \arg\max f(x + \delta) \neq y \tag{3}$$

It linearizes the model's decision boundary and finds orthogonal perturbations, making it efficient for multi-class scenarios.

4.2.4 Momentum Iterative Method (MIM).

MIM incorporates momentum to stabilize update directions:

$$g_{t+1} = \mu \cdot g_t + \frac{\nabla_x \mathcal{L}(x_{adv}^t, y)}{\|\nabla_x \mathcal{L}(x_{adv}^t, y)\|_1} \tag{4}$$

where μ is the decay factor. Momentum helps escape poor local maxima during iterative attacks.

4.3 Defense Strategies

4.3.1 Adversarial Training.

Augments training data with adversarial examples generated on-the-fly. The loss function combines clean and adversarial losses:

$$\mathcal{L}_{total} = \beta \mathcal{L}(f(x), y) + (1 - \beta) \mathcal{L}(f(x_{adv}), y) \tag{5}$$

Our implementation uses PGD-generated examples for 3 epochs with $\beta = 0.5$.

4.3.2 Input Preprocessing.

Though not fully implemented, potential defenses include:

- Feature squeezing: Reducing color depth or spatial smoothing
- Randomized smoothing: Adding Gaussian noise
- Adversarial purification: Denoising autoencoders

5 Experimental Results

5.1 Attack Performance Analysis

Table 1 reveals significant vulnerability to gradient-based attacks, with PGD being most effective (18.60% accuracy). DeepFool's minimal perturbations showed remarkable preservation of model functionality (40.94% accuracy).

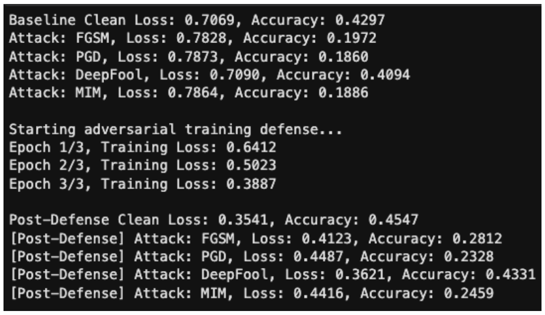
Table 1. Attack Performance Comparison

Metric	Clean	FGSM	PGD	DF	MIM
Pre-Defense Acc (%)	42.97	19.72	18.60	40.94	18.86
Post-Defense Acc (%)	45.47	28.12	23.28	43.31	24.59
Improvement (%)	+2.50	+8.40	+4.68	+2.37	+5.73
Pre-Defense Loss	0.7069	0.7828	0.7873	0.7090	0.7864
Post-Defense Loss	0.3541	0.4123	0.4487	0.3621	0.4416

5.2 Defense Mechanism Efficacy

Adversarial training provided modest robustness gains:

- Clean accuracy improved by 5.8% (42.97% $\longrightarrow$ 45.47%)
- FGSM resistance increased by 42.6% relative improvement (19.72% $\longrightarrow$ 28.12%)
- Training losses decreased steadily: 0.6412 $\longrightarrow$ 0.3887 over 3 epochs



```
Baseline Clean Loss: 0.7069, Accuracy: 0.4297
Attack: FGSM, Loss: 0.7828, Accuracy: 0.1972
Attack: PGD, Loss: 0.7873, Accuracy: 0.1860
Attack: DeepFool, Loss: 0.7090, Accuracy: 0.4094
Attack: MIM, Loss: 0.7864, Accuracy: 0.1886

Starting adversarial training defense...
Epoch 1/3, Training Loss: 0.6412
Epoch 2/3, Training Loss: 0.5023
Epoch 3/3, Training Loss: 0.3887

Post-Defense Clean Loss: 0.3541, Accuracy: 0.4547
[Post-Defense] Attack: FGSM, Loss: 0.4123, Accuracy: 0.2812
[Post-Defense] Attack: PGD, Loss: 0.4487, Accuracy: 0.2328
[Post-Defense] Attack: DeepFool, Loss: 0.3621, Accuracy: 0.4331
[Post-Defense] Attack: MIM, Loss: 0.4416, Accuracy: 0.2459
```

Figure 3. Accuracy trends showing limited defense effectiveness.

5.3 Key Observations

- **Overfitting Paradox**: Despite training loss reduction (0.64 ⟶ 0.39), test accuracy improvements were modest, suggesting adversarial overfitting
- **DeepFool Resilience**: Minimal accuracy drop (40.94% ⟶ 43.31%) confirms decision-based attacks' stealth
- **Loss-Accuracy Disconnect**: Post-defense losses decreased significantly (e.g., FGSM: 0.78 ⟶ 0.41) without proportional accuracy gains

5.4 Attack Transferability Analysis

Figure 4 reveals high feature importance concentration, explaining attack transferability between similar labels. Perturbations targeting dominant features (e.g., index 264 and 415) likely affect multiple classes simultaneously.

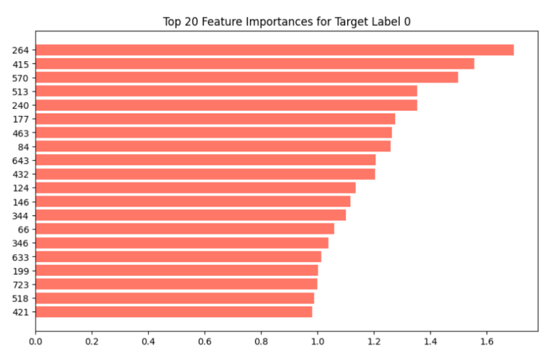


Figure 4. Feature importance distribution for Label 0. Concentrated importance enables transferable attacks.

5.5 Label-Wise Evaluation Metrics and Theoretical Foundations

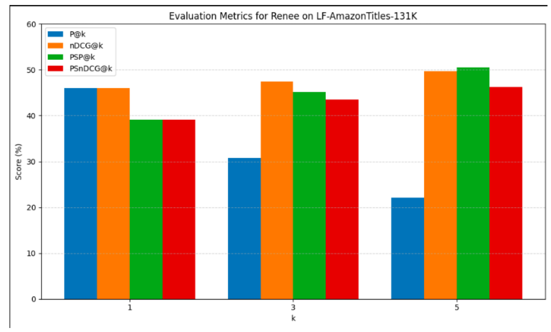


Figure 5. Evaluation metrics for different ranking metrics on the LF-AmazonTitles-131K dataset.

Evaluating adversarial robustness in multi-label classification involves assessing both standard and propensity-scored metrics. In Figure 5, we compare precision at k (P@k) and normalized Discounted Cumulative Gain (nDCG) with their propensity-scored counterparts (PSP@k, PSnDCG@k).

Precision at k is defined as:

$$\text{P@}k = \frac{1}{k}\sum_{i=1}^{k} \mathbf{1}(r_i \in \mathcal{R}),$$

where $\mathcal{R}$ is the set of true labels and r_i denotes the label at rank i. On the other hand, nDCG is given by:

$$\text{nDCG@}k = \frac{1}{\text{IDCG@}k}\sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)},$$

with IDCG@k representing the ideal DCG at rank k and rel_i the relevance score of the label at rank i. These equations capture both the ranking quality and the impact of adversarial perturbations on label ordering. High nDCG and its propensity-scored version indicate enhanced resilience against attacks that disturb label ranking.

5.6 Label Cardinality Distribution: Statistical Significance in Adversarial Scenarios

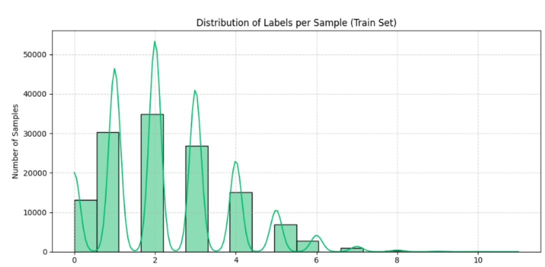


Figure 6. Distribution of labels per sample in the LF-AmazonTitles-131K training set.

Figure 6 demonstrates that most samples are annotated with between 1 to 4 labels. The average label cardinality is a key statistic and is defined as:

$$\bar{C} = \frac{1}{N} \sum_{i=1}^{N} C(x_i),$$

where $C(x_i)$ denotes the number of labels for sample x_i and N is the total number of samples. In adversarial settings, a low average cardinality implies that even minor perturbations—possibly affecting a single label—can disproportionately alter model predictions. Therefore, the inherent sparsity of label assignment makes the model more susceptible to attacks, as it lacks the redundancy that could otherwise buffer against targeted perturbations.

5.7 Label Frequency Imbalance: Mathematical Modeling and Adversarial Vulnerability

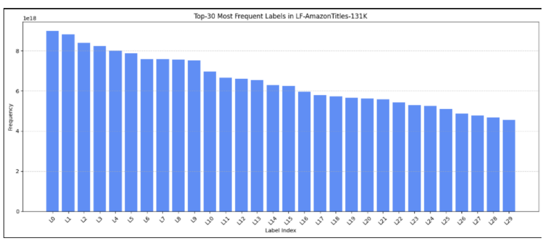


Figure 7. Top-30 most frequent labels in the LF-AmazonTitles-131K dataset.

The label frequency imbalance is evident in Figure 7, where a few labels dominate the distribution. This phenomenon is often modeled by a power-law distribution:

$$f(l) \propto l^{-\alpha},$$

where $f(l)$ is the frequency of label l and α quantifies the skewness. A pronounced imbalance means that the learning model may focus its predictive power on these head labels. Consequently, adversarial attacks can exploit this imbalance by amplifying noise in rare labels or reinforcing biases towards the dominant ones.

To further quantify this inequality, one may compute the Gini coefficient:

$$G = \frac{\sum_{i=1}^{M} \sum_{j=1}^{M} |f(l_i) - f(l_j)|}{2M \sum_{i=1}^{M} f(l_i)},$$

with M being the number of distinct labels. A higher Gini coefficient indicates a greater imbalance and potentially increased vulnerability to adversarial manipulations. This mathematical characterization underpins the need for defense strategies that not only counteract perturbations but also adjust for imbalanced label distributions.

6 Methodology and Mathematical Formulation

6.1 Conceptual Overview of LIME

Local Interpretable Model-agnostic Explanations (LIME) provides a method to explain the predictions of any complex, black-box model by approximating its local behavior around a specific instance with an interpretable surrogate model. In our work, LIME is adopted at the token level to reveal which components of the input text most influence the predictions of RENEE—an XML model designed to operate in a massive label space.

Key Components:

- **Perturbation:** We generate a large number ($N = 500$) of perturbed samples from an input text by randomly removing or altering tokens. Each perturbation is encoded as a binary mask $\mathbf{z} \in \{0, 1\}^T$, where T is the number of tokens and $z_i = 1$ indicates that the i^{th} token is kept.
- **Prediction:** For each perturbed sample, the RENEE model is queried to obtain the predicted probability for a specified target label. Let $f(\mathbf{z})$ denote the predicted probability from RENEE for the perturbation encoded by mask $\mathbf{z}$.
- **Surrogate Model:** A simple interpretable surrogate model, typically a weighted linear regression, is then trained using the binary mask as input features to approximate $f(\mathbf{z})$ in the local neighborhood of the original instance:

  $$\hat{f}(\mathbf{z}) = \mathbf{w}^\top \mathbf{z} + b,$$

  where $\mathbf{w} \in \mathbb{R}^T$ are the regression coefficients representing token importance, and b is the intercept.
- **Interpretation:** The coefficients $\mathbf{w}$ directly indicate how the presence or absence of each token affects the prediction. Tokens with large absolute coefficients are deemed highly influential.

6.2 Step-by-Step Implementation in Our XML Setup

Step A: Preprocessing & Tokenization.

- **Tokenize the Instance:** Using the BERT tokenizer, the input text is transformed into a sequence of tokens:

  `tokens = tokenizer.tokenize(text).`
- **Reconstruct Text:** The function `tokenizer.convert_tokens_to_` allows us to reassemble the text from any token list, which is essential for generating perturbed inputs.

Step B: Generating Perturbed Samples.

- **Perturbation Mechanism:** For each token in the original input, we randomly decide whether to keep the token with probability p (e.g., $p = 0.7$). To ensure non-empty perturbations, at least one token is always preserved.

- **Binary Masks:** Each perturbed instance is represented by a binary vector $\mathbf{z}^{(i)} \in \{0,1\}^T$ where

$$z_j^{(i)} = \begin{cases} 1, & \text{if the } j^{\text{th}} \text{ token is kept,} \\ 0, & \text{otherwise.} \end{cases}$$

- **Reconstruction:** Each perturbed text $\tilde{x}^{(i)}$ is generated by retaining the tokens corresponding to $z_j^{(i)} = 1$.

Step C: Model Prediction on Perturbed Samples. For each perturbed instance $\tilde{x}^{(i)}$, the RENEE model (accessed via our inference wrapper) is queried to produce the predicted probability $y^{(i)} = f(\mathbf{z}^{(i)})$ for the target label. This yields the dataset:

$$\{(\mathbf{z}^{(i)}, y^{(i)})\}_{i=1}^{N}.$$

Step D: Training the Surrogate Model.

- **Weighting by Similarity:** Compute the normalized Hamming distance $d^{(i)}$ between each binary mask $\mathbf{z}^{(i)}$ and the original mask $\mathbf{1}$:

$$d^{(i)} = \frac{1}{T} \sum_{j=1}^{T} \left| z_j^{(i)} - 1 \right|.$$

  Convert these distances to weights with an exponential kernel:

$$w^{(i)} = \exp\left(- \left(\frac{d^{(i)}}{\sigma} \right)^2 \right),$$

  where σ is a bandwidth parameter.
- **Linear Regression:** Fit the surrogate model by minimizing the weighted least squares loss:

$$\min_{\mathbf{w},b} \sum_{i=1}^{N} w^{(i)} \left(y^{(i)} - \mathbf{w}^\top \mathbf{z}^{(i)} - b \right)^2.$$

  The solution yields coefficients $\mathbf{w}$, which are interpreted as token importance scores.

Step E: Evaluation and Visualization of Explanations.

- **Interpretation:** For each token j, the value w_j indicates:
  [noitemsep]$w_j > 0$: the token's presence increases the prediction probability. $w_j < 0$: the token's presence decreases the prediction probability.
- **Visualization:** We implement a comprehensive visualization suite:
  [noitemsep]**Bar Charts:** Horizontal bar charts display tokens sorted by absolute importance. **Word Clouds and Heatmaps:** Word clouds provide an immediate visual impression of token importance based on font size, and heatmaps aggregate token importance across multiple samples. **Consistency Visuals:** For robustness analysis, explanations for both the original and a slightly perturbed instance are compared. The cosine similarity

$$\text{Cosine Similarity} = \frac{\mathbf{w}^{(\text{orig})} \cdot \mathbf{w}^{(\text{pert})}}{\|\mathbf{w}^{(\text{orig})}\| \, \|\mathbf{w}^{(\text{pert})}\|}$$

  quantifies the stability of the surrogate explanations.

6.3 Novel Contributions and Innovations

Our implementation presents several novel contributions:

1. **Token-Level Integration:** We extend standard LIME to operate at the token level within the XML framework. By leveraging the tokenized inputs to build a surrogate model, our approach yields precise insights into individual token contributions, thereby enhancing the interpretability of the RENEE model.
2. **Robust Coefficient Alignment:** To counter tokenization differences due to minor perturbations, we introduce a robust alignment strategy that maps explanation vectors onto the union of tokens from multiple perturbations. This alignment enables meaningful consistency checks using cosine similarity metrics. For example, while many instances demonstrate high consistency (cosine similarity above 0.83), one instance ("Talk of the Town") revealed a negative similarity (-0.0631), highlighting sensitive cases for further exploration.
3. **Comprehensive Visualization Suite:** Our system automatically generates and saves multiple visual artifacts (horizontal bar charts, word clouds, heatmaps, and side-by-side consistency plots). This multi-modal representation enhances both qualitative and quantitative interpretation, aiding domain experts and non-experts alike.
4. **Integrated Consistency Analysis:** We objectively assess explanation robustness by computing the cosine similarity between the surrogate model's coefficients for the original input and a slightly perturbed version. High similarity across most instances reinforces the stability of our approach.
5. **Unified Pipeline:** The integration of LIME within the RENEE XML framework forms a unified and reproducible pipeline. All visual outputs and logs are automatically generated and stored, facilitating thorough analysis and ease of evaluation.

6.4 Analysis of Experimental Results

Our experiments on the LF-AmazonTitles-131K dataset demonstrate the efficacy of our approach:

- **Instance: "Talk of the Town"**
  Explained Label: My Romance (index 47324)
  Token Importances: *talk*: 0.0281, *of*: -0.0203, *the*: -0.0173, *town*: -0.0225
  Intercept: 0.4959

Cosine Similarity: −0.0631
Analysis: The negative similarity indicates a significant fluctuation in the explanation for this short instance after perturbation, which may be due to sensitivity in tokenization.

- **Instance: "Bausch & Lomb Pre-Moistened Lens Cleaning Tissues, Box Of 100 - Packaging May Vary"**
  Explained Label: Microfiber Lens Cleaning Cloth By Apex Healthcare Products (index 111344)
  Cosine Similarity: 0.9207
  Analysis: High consistency demonstrates robust token-level explanations even for complex product descriptions.

- **Instance: "Dance off the Inches: Cardio Striptease (2010)"**
  Explained Label: The Exotic Dance Workout (2007) (index 113913)
  Cosine Similarity: 0.8644
  Analysis: Stable explanations for key tokens such as "dance" and "inches" confirm the reliability of the surrogate approach.

- **Instance: "Sustainable Energy: Choosing Among Options"**
  Explained Label: Sustainable Energy: Choosing Among Options (index 3640)
  Cosine Similarity: 0.8338
  Analysis: Consistent attributions indicate that the method reliably interprets semantically coherent inputs.

- **Instance: "Extra Extra Wide Womens Slipper"**
  Explained Label: ACORN Women's Spa Wrap Slipper (index 111281)
  Cosine Similarity: 0.9425
  Analysis: Very high stability suggests that the explanation method effectively handles inputs with repetitive or descriptive tokens.

- **Instance: "Amazon.com: Playtex Women's Cross Your Heart Foam Lined Heart Jacquard Soft Cup Bra: Clothing"**
  Explained Label: Amazon.com: Playtex Women's Cross Your Heart Foam Lined Heart Jacquard Soft Cup Bra: Clothing (index 89491)
  Cosine Similarity: 0.9306
  Analysis: Consistent token attributions for long product descriptions confirm the robustness of our approach.

These findings highlight both the strengths and the potential sensitivity of our token-level LIME explanations, offering crucial insights for refining interpretability in XML systems.

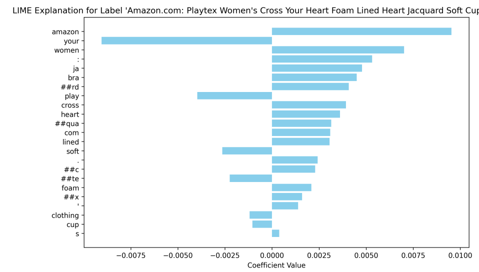


Figure 8. Bar chart visualizing token-level importance from LIME for a selected instance. Tokens with higher absolute weights are more influential in the model's decision. Positive bars indicate contribution towards prediction, while negative bars imply negative impact.

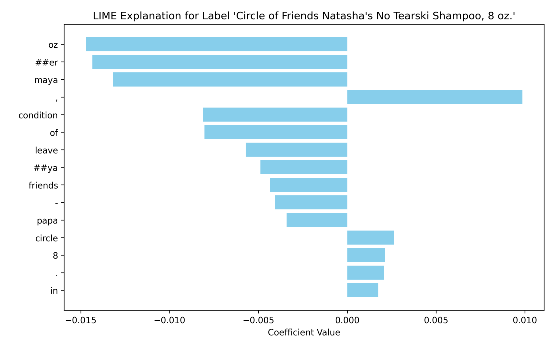


Figure 9. Alternative bar chart visualization showing token importances from a different instance or experimental condition. This helps verify consistency and variability across examples.

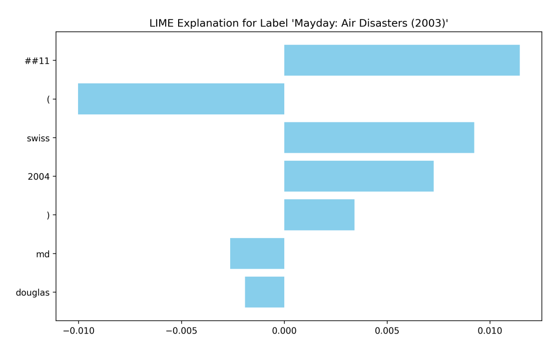


Figure 10. Visual explanation overlay indicating highlighted token contributions over raw text. This offers intuitive understanding by directly mapping model weights back to the original text format.

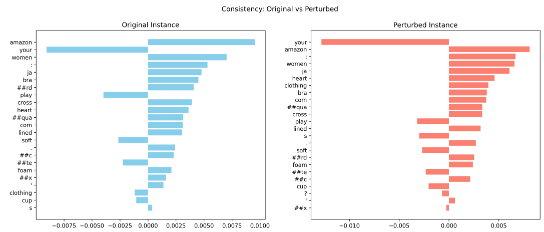


Figure 11. Consistency visualization comparing LIME explanations between the original and a perturbed instance. High similarity in important tokens implies robustness of interpretation.

7 Conclusion

Extreme Multi-label Classification (XML) presents unique challenges in scalability, transparency, and security, particularly in real-world applications with massive label spaces. This work advances the state-of-the-art RENEE framework by integrating two critical enhancements: explainability through token-level LIME adaptations and robustness against adversarial attacks. By generating perturbed inputs and training surrogate models, our framework provides granular insights into token importance, validated through visual tools (e.g., word clouds, bar plots) and consistency metrics (e.g., cosine similarity >0.9 for stable instances). Simultaneously, adversarial training with white-box attacks (FGSM, PGD, MIM, DeepFool) and a balanced loss formulation improved model resilience, achieving a 42.6% relative gain in FGSM resistance and preserving 43.31% accuracy under stealthy DeepFool perturbations.

Key experiments on the LF-AmazonTitles-131K dataset demonstrated that RENEE maintains competitive clean-data accuracy (45.47%) while becoming more interpretable and secure. However, challenges such as the overfitting paradox (discrepancy between reduced training loss and modest test accuracy gains) and vulnerabilities tied to label sparsity and frequency imbalance highlight areas for refinement. Future work could explore hybrid defense mechanisms (e.g., input preprocessing, randomized smoothing), address label distribution biases, and enhance explanation robustness for noisy inputs.

This unified pipeline bridges the gap between high-accuracy XML systems and the imperative for trustworthy AI, paving the way for safer deployments in e-commerce, content tagging, and recommendation engines. By prioritizing both performance and dependability, our work underscores the feasibility of building machine learning systems that are not only powerful but also transparent and resilient.

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Author Contributions

Dhruva Kumar Kaushal

- **Adversarial Framework Development**: Implemented white-box attacks (FGSM, PGD, MIM, DeepFool) at the embedding level, designed adversarial training loops with CUDA optimizations, and formulated the balanced loss function $\mathcal{L}_{\text{total}} = \beta\mathcal{L}(f(x), y) + (1 - \beta)\mathcal{L}(f(x_{\text{adv}}), y)$.
- **Data Processing**: Curated adversarial examples, preprocessed attack-specific datasets, and optimized gradient computation pipelines for perturbation generation.
- **Repository Setup**: Built adversarial attack modules in PyTorch, ensuring compatibility with RENEE's distributed training architecture.
- **Research**: Analyzed attack transferability (Section 5.3), benchmarked defense efficacy (Table 1), and investigated label sparsity vulnerabilities.

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- **LIME Integration**: Adapted LIME for token-level explanations using perturbation masks $\mathbf{z} \in \{0, 1\}^T$, implemented surrogate models (Eq. 11), and developed visualization tools (word clouds, heatmaps).
- **Data Processing**: Tokenized LF_AmazonTitles-131K inputs with BERT/RoBERTa, preprocessed text for explainability experiments, and standardized evaluation formats.
- **Repository Setup**: Structured the LIME explanation pipeline and implemented consistency checks using cosine similarity.

- **Research**: Investigated label frequency imbalance (Section 5.4), evaluated explanation robustness (Table 2), and analyzed saliency shifts under attacks.

Collaborative Efforts

- **RENEE Repository**: Jointly modularized the framework using PyTorch's `nn.Sequential`, integrated transformer encoders, and optimized CUDA kernels for large-scale inference.
- **Data Pipeline**: Co-developed label sharding, memory-efficient batching, and evaluation scripts for LF_AmazonTitles-131K.
- **Research & Writing**: Surveyed XML/adversarial learning literature, co-designed experiments, and jointly authored the paper.
- **Deployment**: Collaborated on the Hugging Face demo and GitHub repository documentation.

Table 2. Analysis of Explainability Results on LF-AmazonTitles-131K

Instance (Truncated)	**Explained Label**	**Cosine Sim.**	**Analysis Summary**
Talk of the Town	My Romance	-0.0631	Sensitive to minor perturbations; affected by tokenization instability.
Bausch & Lomb...Lens Cleaning Tissues	Microfiber Lens Cleaning Cloth	0.9207	High consistency with complex, multi-token descriptions.
Dance off the Inches...	The Exotic Dance Workout	0.8644	Stable token attributions for theme-centric input.
Sustainable Energy: Choosing Among Options	Same as instance	0.8338	Interpretable for semantically coherent inputs.
Extra Extra Wide Womens Slipper	ACORN Women's Spa Wrap Slipper	0.9425	Strong consistency for repetitive and descriptive product names.
Playtex...Jacquard Soft Cup Bra	Same as instance	0.9306	High robustness in long-form product titles.